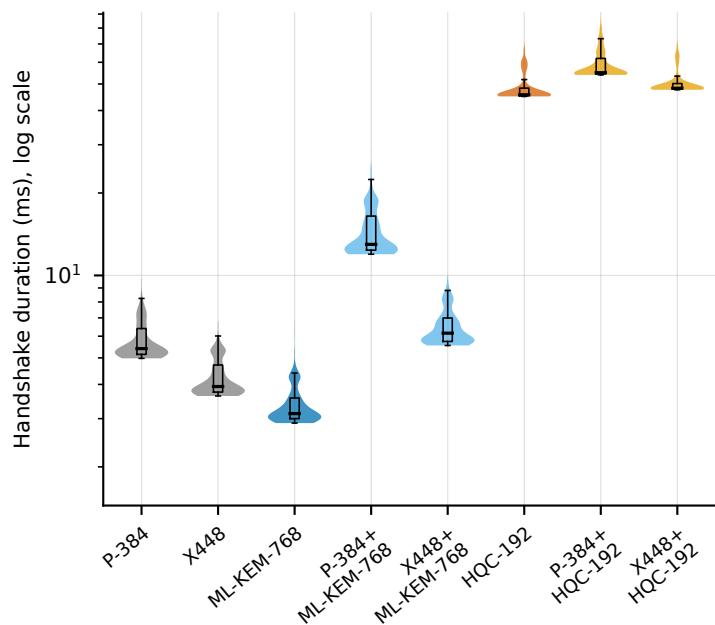
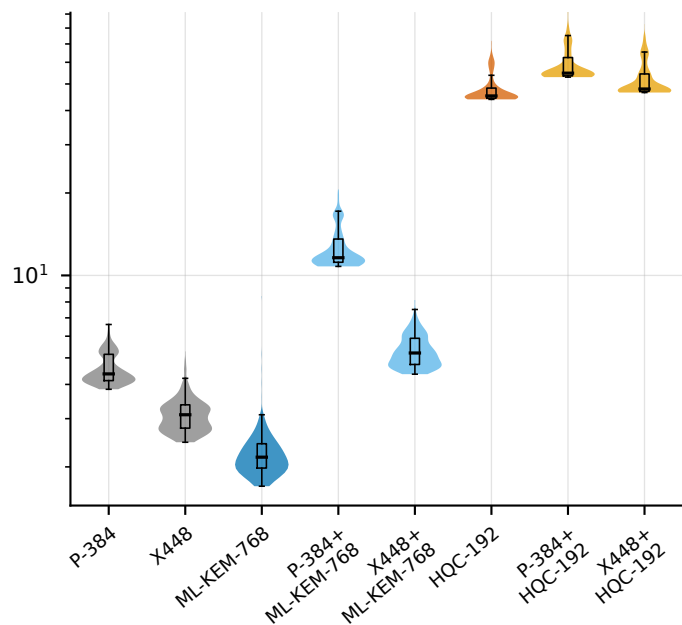


(a)

secp384r1 (classical)

**(b)**

mlDSA65 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp384r1 Mean (ms)	secp384r1 CV (%)	secp384r1 Max (ms)	mlDSA65 Mean (ms)	mlDSA65 CV (%)	mlDSA65 Max (ms)
P-384	5.84	15.9	9.74	4.60	12.9	6.62
X448	4.28	15.8	6.80	3.14	14.5	5.28
ML-KEM-768	3.39	16.4	6.85	2.25	20.4	8.40
P-384+ML-KEM-768	14.6	20.1	29.1	12.7	17.3	20.6
X448+ML-KEM-768	6.54	15.0	9.97	5.37	13.6	8.13
HQC-192	48.6	11.5	71.4	47.9	11.9	71.9
P-384+HQC-192	59.1	12.1	93.7	59.1	13.8	96.2
X448+HQC-192	50.8	10.4	71.5	52.2	14.6	82.7